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1. Drawing General

Check area	Review prompt	Applicability
Title block	Confirm drawing number, title and document description are correct.	All drawings
Title block	Confirm revision and revision date are current and consistent with the issue.	All drawings
Title block	Confirm project, client and Maxwell title block details are correct, including required client logos.	All drawings
Drawing layout	Confirm sheet size, scale/orientation and plotted extents are appropriate and nothing is clipped.	All drawings
Drawing layout	Confirm line numbers / grid references are present where required and follow the project convention.	Where applicable
Drawing layout	Confirm drawing references, continuation references and sheet-to-sheet links are correct.	Where applicable
Drawing content	Confirm equipment tags, descriptions and identifiers are consistent with the rest of the drawing set.	All drawings
Drawing content	Confirm internal and external wiring/cabling is represented using the correct line types and conventions.	Where applicable
Presentation	Check for overlapping text, orphaned notes, broken leaders, duplicated objects or other plotting artefacts.	All drawings
Presentation	Confirm notes, legends, symbols and abbreviations are current and relevant to this drawing.	All drawings

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2. Drawing Motor / Pump Schematic

Check area	Review prompt	Applicability
Equipment details	Confirm motor/pump tag, description and kW rating are correct.	All motor/pump schematics
Protection	Confirm incoming protective device tag/label matches the Single Line Diagram.	All motor/pump schematics
Starter / drive	Confirm starter, contactor, overload, VSD or soft starter part numbers are correct and suitably rated for the motor.	As applicable
Wiring	Confirm line numbers and wire numbers are correct and updated for this drawing.	All motor/pump schematics
I/O	Confirm PLC I/O allocations and signal descriptions agree with the I/O schedule and I/O drawings.	Where PLC controlled
Indication	Confirm required status indications are provided in accordance with the client standard, e.g. ON, OFF, RUN, AVAILABLE and FAULT.	As required by client
Control	Review the complete control philosophy and confirm the circuit achieves the intended operating sequence and interlocks.	All motor/pump schematics
Motor protection	Confirm motor thermistor / temperature protection is included where required.	Where required
Submersible motors	For submersible motors, confirm seal leakage / MiniCAS or equivalent monitoring requirements are addressed.	Submersible motors only
VSD – rating	Confirm the VSD part number and current/kW rating are suitable for the motor, including any required drive upsizing or derating.	VSD only
VSD – ventilation	Confirm cubicle ventilation is provided where required and the fan control operates with drive operation and/or high temperature as specified.	VSD only
VSD – safety	Confirm the stop / emergency-stop philosophy uses Safe Torque Off or the specified safety function where required.	VSD only
VSD – status	Confirm drive relays / digital outputs are configured for the required fault, run, remote/available and other status signals.	VSD only
VSD – protection	Confirm semiconductor / high-speed line-side fuses are provided where required by the drive manufacturer or design.	VSD only

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VSD – communications	Confirm any required communications expansion module is shown with the correct protocol and part number.	VSD only
VSD – 24 VDC	Confirm any dedicated 24 VDC supply required by the drive or controls is shown.	VSD only
VSD – manual control	Confirm a local potentiometer / manual speed reference is provided where required by the operating philosophy.	VSD only
VSD – location	Confirm internal/external drive location is clear and terminal marshalling reflects the actual installation arrangement.	VSD only
VSD – earthing	Confirm motor cable screening, EMC earthing and dedicated drive earth requirements are detailed correctly.	VSD only
DOL – control	Confirm the control circuit fully implements start, stop, permissives, trips and required interlocks.	DOL only
DOL – reversing	Confirm reversing contactors and electrical/mechanical interlocking are provided where reversing is required.	DOL reversing only
DOL – emergency stop	Confirm emergency-stop operation removes or disables the control circuit as required by the design.	DOL only
DOL – overload	Confirm thermal overload protection is provided and suitably rated.	DOL only
DOL – latching	Confirm latching / seal-in logic is correct for the required operating mode, including forward/reverse arrangements.	DOL only
Soft starter – bypass	Confirm internal or external bypass arrangement is correctly shown and coordinated with the selected soft starter.	Soft starter only
Soft starter – part number	Confirm the soft starter part number and rating are shown and suitable for the motor.	Soft starter only
Soft starter – ventilation	Confirm cubicle ventilation is provided where required and controlled appropriately.	Soft starter only
Soft starter – status	Confirm fault and other required status feedbacks are captured.	Soft starter only
Soft starter – protection	Confirm semiconductor / high-speed fuses are provided where required.	Soft starter only

3. Single Line Diagram

Check area	Review prompt	Applicability
Supply / transformer	Confirm transformer ratings and key details are shown correctly, including kVA rating.	Where applicable
Earthing system	Confirm the earthing arrangement is correctly identified (e.g. TN-C-S) and the MEN connection is shown at the correct location.	All applicable LV systems
Metering	Confirm supply-authority and/or private metering requirements are captured and the metering method is correct, e.g. CT-operated or direct connected.	Where metering required
Loads	Confirm all loads are shown with correct tags, descriptions and design/load currents where required.	All SLDs
Protection	Confirm protective device tags, ratings and settings are correctly labelled and coordinated with the design.	All SLDs
Changeover	Confirm ATS / MTS requirements are shown and the arrangement reflects the intended operating philosophy.	Where applicable
Generator	Confirm generator connection and kVA rating are shown. For temporary generators, state the minimum required rating where applicable.	Where applicable
Phase monitoring	Confirm phase-failure / voltage-monitoring relays are provided at the required normal, alternate and load-side locations.	Where required
Harmonics	Confirm any active harmonic filter is shown and associated CT locations / directions are accounted for.	Where applicable
Continuations	Where the SLD spans multiple sheets, confirm continuation references and previous/next drawing references are correct.	Multi-sheet SLDs
Cabling	Confirm internal and external cables are identified correctly and use the correct line type / notation.	All SLDs
Surge protection	Confirm surge protection is included where required and the SPD type/location is suitable for the installation, including Type 1+2 at the incomer where specified.	Where required

4. 24VDC Distribution Drawing

Check area	Review prompt	Applicability
Supply / transformer	Confirm transformer ratings and key details are shown correctly, including kVA rating.	Where applicable
Architecture	Confirm the required UPS / DC supply arrangement is shown and provides the specified level of redundancy.	All 24 VDC systems
Power supplies	Confirm power supply ratings and quantities are adequate for the connected load and required redundancy.	All 24 VDC systems
Surge protection	Confirm surge protection is provided on the AC/DC supply arrangement where required.	Where required
Redundancy module	Confirm any redundancy module is required, correctly selected and connected in accordance with the manufacturer arrangement.	Redundant supplies only
Ride-through	Where AC UPS changeover or short interruptions can occur, confirm a buffer / capacitor module is provided if required.	Where required
Distribution	Confirm all motor-control devices, PLC/network equipment, instruments and other 24 VDC loads are accounted for.	All 24 VDC systems
Protection	Confirm individual distribution protective devices are appropriately rated, labelled and allocated.	All 24 VDC systems
References	Confirm device tags and circuit references align with associated schematics, network drawings and I/O drawings.	All 24 VDC systems

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5. Network Drawing

Check area	Review prompt	Applicability
PLC hardware	Confirm PLC model, rack size, slot allocation and card part numbers align with the I/O schedule.	All PLC networks
PLC communications	Confirm required Ethernet / communications modules are shown with correct part numbers and protocols.	Where applicable
HMI	Confirm local and external HMIs are shown, tagged and connected correctly.	Where applicable
Radio	Confirm radios and antennas are shown and surge protection is included where required.	Radio systems only
Remote I/O	Confirm remote I/O stations and interconnections are shown, including power and network links.	Remote I/O only
Fibre	Confirm fibre links identify fibre type/core requirements, termination hardware (e.g. FOBOT) and required media/protocol converters.	Fibre links only
Protocol	Confirm communication protocols are explicitly identified and consistent across connected equipment.	All networks
Topology	Confirm switch ports, network paths and redundant/ring topology are represented correctly where applicable.	As applicable
External interfaces	Confirm interfaces leaving the assembly are clearly distinguished from internal network wiring.	Where applicable

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6. Earthing Drawing

Check area	Review prompt	Applicability
Conductor sizes	Confirm actual protective earthing conductor sizes are stated, not only associated cable sizes.	All earthing drawings
Lightning protection	Confirm the lightning protection system and connections are detailed where required.	Where LPS provided
VSD earthing	Confirm dedicated VSD earthing, motor PE and screen termination requirements are shown.	Where VSDs provided
Field earthing	Confirm allowances are provided for field equipment earthing where required.	Where applicable
Distribution earths	Confirm earth conductors to field distribution boards / areas are appropriately sized and shown.	Where applicable
Lightning earth network	Confirm lightning down conductors / earth termination network are arranged so lightning current is not intentionally routed through the switchboard assembly.	Where applicable
MEN and electrodes	Confirm MEN connection, earth electrodes and electrode locations/quantities are shown where applicable.	Where applicable
Cable schedule	Confirm associated cable tags and internal/external earthing cables agree with the cable schedule.	Where scheduled
Notes	Confirm special earthing, bonding and screen termination requirements are stated in drawing notes.	Where required
Sizing reference	Confirm the drawing references the relevant calculation / PSA for minimum earthing conductor sizing where required.	Where calculated

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7. IO Drawing

Check area	Review prompt	Applicability
PLC allocation	Confirm PLC, rack, slot, card and channel details are correct and align with the I/O schedule.	All I/O drawings
24 VDC supply	Confirm the 24 VDC distribution circuit / CB reference is correct.	Where applicable
Instrument wiring	Confirm instruments are represented with the correct 1-, 2-, 3- or 4-wire arrangement.	Instrument I/O
Equipment tags	Confirm equipment tags exist in the relevant valve, instrument, mechanical or process equipment schedules.	All I/O equipment
Duplicate detail	Confirm equipment is fully detailed only once; where shown elsewhere, use a clear drawing reference rather than duplicating the circuit.	Where cross-referenced
Cross references	Confirm wire numbers, slot/channel references and external drawing references agree with connected drawings.	All cross-referenced I/O
Continuity	Trace each signal through referenced drawings and confirm continuity of tag, terminal, wire/core and signal description.	All cross-referenced I/O
Signal type	Confirm digital/analog signal type, polarity and normal state are represented correctly.	As applicable

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8. Incomer Schematic

Check area	Review prompt	Applicability
SLD alignment	Confirm the incomer schematic reflects the Single Line Diagram topology and equipment arrangement.	All incomer schematics
Metering	Confirm required metering is captured and the CT / direct-connected arrangement is correct.	Where metering required
Changeover	Confirm ATS / MTS controls and interfaces are captured where applicable.	Where applicable
Surge protection	Confirm surge protection devices and associated protection/monitoring are shown.	Where required
Phase monitoring	Confirm phase-failure / voltage-monitoring relays are captured in the correct locations.	Where required
Outgoing circuits	Confirm any main outgoing circuits required on the incomer schematic are captured and consistent with the SLD.	Project dependent
Protection settings	Confirm detailed protective-device settings are omitted from the schematic where the project convention requires them to appear only on the SLD.	Where project convention applies
Tags	Confirm incomer CB, ATS/MTS and other equipment tags align with the SLD and schedules.	All incomer schematics
Control power	Confirm control supplies, trip/close circuits and status feedbacks are complete and correctly referenced.	Where applicable

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9. Schedule General

Check area	Review prompt	Applicability
Header	Confirm project number, client, document title, revision, date and other required header information are correct.	All schedules
Template	Confirm the latest approved Maxwell/client template is being used.	All schedules
Revision	Confirm revision information and change notes are current and consistent with the issue.	All schedules
Plot / PDF	Confirm the workbook plots correctly to PDF with no clipped columns, blank pages, broken print areas or other artefacts.	All schedules
Presentation	Check fonts, row heights, column widths, page breaks and repeated headers are consistent and readable.	All schedules
Completeness	Confirm there are no hidden filters, unintended hidden rows/columns, stale notes or temporary working data in the issue copy.	All schedules

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10. Cable Schedule

Check area	Review prompt	Applicability
Incoming supply	Confirm incoming supply cables are included, including neutral and protective earth conductors as applicable.	All projects
Generator	Confirm generator power, battery charger, communications and control cables are included where applicable.	Generator projects
Motor power	Confirm all motor power cables are included, including multiple cable sections created by external VSDs, field LCSs or other intermediate equipment.	Motor projects
Networks	Confirm all network cabling leaving the assembly is included, including Ethernet, fibre, RS-485 and other field communications.	Where applicable
Junction boxes	Confirm main cables to junction boxes and all outgoing field cables from each junction box are captured.	Where applicable
Analog cable type	Confirm analog/instrument cable screening type matches the client specification, including individual vs overall screening requirements.	Analog/instrument cables
Lengths	Confirm cable lengths are included where required and are reasonable for the installation.	Where required
Drawing references	Confirm motor power cable references align with motor schematics and I/O cable references align with I/O drawings.	All referenced cables
Multiple drawing refs	Where one cable relates to multiple drawings, use the project convention (e.g. "Varies") rather than an incomplete list of references.	Where applicable

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11. IO Schedule

Check area	Review prompt	Applicability
Equipment coverage	Confirm all I/O equipment exists in the relevant valve, instrument, mechanical or process equipment schedule, or is otherwise clearly identified in the drawing set.	All I/O
Special devices	Confirm incomer phase-failure relays and special protection/monitoring devices requiring I/O are included.	Where applicable
Motor signals	Confirm every motor/pump schematic is represented and required commands/status feedbacks match the schematic.	Motor projects
Signal naming	Confirm signal descriptions are consistent between drawings and schedule, e.g. FLOW, FLOW LOW, RUN, FAULT.	All I/O
Valves	Confirm each valve has the correct command/feedback set for its actual functionality, including spring return, open/closed limits and analog position where required.	Valve I/O
Analog signals	Confirm analog input/output ranges, engineering quantity and signal type are correct where shown.	Analog I/O
PLC allocation	Confirm PLC, rack, slot, channel and terminal allocations agree with the network and I/O drawings.	All allocated I/O
Uniqueness	Confirm duplicate channels, duplicate tags and unintended spare/used conflicts are not present.	All I/O
Spares	Confirm spare I/O allowance meets project/client requirements and is clearly identified.	Where required

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12. Report General

Check area	Review prompt	Applicability
Front matter	Confirm report title, project, client, document number, revision, date, author, reviewer and approver details are correct.	All reports
Revision	Confirm revision history and issue description reflect the current issue.	All reports
Scope	Confirm scope, exclusions, assumptions and limitations accurately reflect the work undertaken.	All reports
References	Confirm standards, drawings, calculations and source documents are current and correctly referenced.	All reports
Internal consistency	Confirm figures, tables, section references, equipment tags and stated values are consistent throughout the report.	All reports
Conclusions	Confirm conclusions and recommendations are supported by the analysis and clearly identify any outstanding actions.	All reports
PDF navigation	Confirm the PDF contains usable bookmarks for major sections and appendices, with working internal links where provided.	All reports
PDF quality	Check the PDF for blank pages, clipping, broken figures, missing fonts, low-resolution graphics and other export artefacts.	All reports
Markup removal	Confirm comments, tracked changes, review markups and hidden working notes have been removed from the issue PDF.	All reports
Attachments	Confirm appendices and embedded/attached supporting documents are complete and in the correct order.	Where applicable

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13. Calculation General

Check area	Review prompt	Applicability
Identification	Confirm project, calculation title/number, revision, date, author and reviewer details are correct.	All calculations
Purpose	Confirm the calculation clearly states its purpose, design case and acceptance criteria.	All calculations
Inputs	Confirm input data is current, traceable and consistent with drawings, schedules and vendor information.	All calculations
Assumptions	Confirm assumptions are explicit, technically reasonable and conservative where required.	All calculations
Method	Confirm the selected calculation method and equations are appropriate and referenced to the relevant standard/guidance.	All calculations
Units	Confirm units are stated and consistently applied throughout.	All calculations
Results	Check key calculations independently or by reasonableness test and confirm no obvious formula/reference errors are present.	All calculations
Design impact	Confirm calculated results are correctly transferred into drawings, equipment ratings, settings and schedules.	All calculations
Margins	Confirm required design margins, tolerances and derating factors are included.	Where applicable
Output	Confirm the issued PDF is complete, legible and free of comments, hidden worksheets or calculation artefacts.	All calculations

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14. Maximum Demand Calculation

Check area	Review prompt	Applicability
Method	Confirm maximum demand has been determined in accordance with Appendix C / Clause C2 of AS/NZS 3000 where applicable, or that any alternative method is clearly justified.	All maximum-demand calculations
Load list	Confirm all connected loads are included with correct ratings and operating modes.	All maximum-demand calculations
Demand factors	Confirm duty / demand factors are correct for each load category and consistent with the selected calculation method.	All maximum-demand calculations
Diversity	Confirm assumed diversity, standby duty and mutually exclusive loads are clearly identified and technically justified.	Where applicable
Largest motor	Confirm any treatment of largest motor / motor starting contribution required by the chosen method is applied correctly.	Motor installations
Generator capacity	Compare calculated maximum demand with available site/generator capacity and confirm adequate margin.	Where generator/limited supply applies
Mains sizing	Confirm mains and feeder cable sizing is consistent with the calculated design current / maximum demand.	All applicable feeders
Switchgear rating	Confirm incomer and busbar continuous-current ratings are adequate for the calculated demand.	Switchboard calculations
References	Confirm load ratings and assumptions trace back to the current drawings, equipment schedules and vendor data.	All maximum-demand calculations